Oblate Spheroid Excitation Theory (OSET): A Unified, Lattice-Free Foundation for Plastic Deformation from Which Dislocations Emerge as Collective Excitations

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Abstract

Classical plasticity theory takes the dislocation as its irreducible elementary entity. This fails for grain boundaries, amorphous solids, nanocrystals near the glass transition, ceramics, and polymers because the dislocation is a topological concept that requires a periodic crystal lattice for its very definition. We propose the *Oblate Spheroid Excitation Theory* (OSET), in which the elementary excitation is a shear-eigenstrained oblate spheroid (the oblate-spheroidal transformation zone, OSTZ) embedded in an elastic continuum. The OSTZ requires no lattice, carries a finite energy $\Delta F_0 = \frac{1}{2}(\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2)GV_0$, and has a non-singular stress field. Three central results are proved: (1) a single OSTZ is equivalent in the far field to a dislocation dipole; (2) a coplanar chain of N OSTZs is mathematically identical to a Peierls–Nabarro dislocation with core width $\zeta = W$ and Burgers vector $b = N\gamma_0 W$; (3) a full lattice dislocation nucleates when $N = N_c = b_{\mathrm{latt}}/(\gamma_0 W)$. OSET naturally predicts the theoretical shear strength, Peierls stress, stacking-fault energy, core energy, and Frank–Read critical stress, all as derived quantities. Since dislocations emerge from OSET but OSET does not require dislocations, OSET is the more fundamental description.

Keywords: oblate spheroid excitation, Eshelby inclusion, dislocation emergence, grain-boundary sliding, Peierls–Nabarro model, superplasticity

1. Introduction

1.1. What dislocation theory requires

The classical dislocation is defined via the Burgers circuit in a Bravais lattice:

$$\oint_C \mathrm{d}u_i = b_i, \qquad b_i \in \Lambda_{\mathrm{Bravais}}. \tag{1}$$

This definition carries three unavoidable restrictions.

(R1) Requires a crystal lattice. No lattice, no Burgers vector, no dislocation. Grain boundaries, glasses, and polymers are excluded by definition.

(R2) Singular core. The Volterra stress $\sigma \sim Gb/2\pi r \to \infty$; a cutoff $r_0 \sim b$ is introduced empirically.

(R3) Athermal. Thermal activation is appended via kink nucleation; it is not intrinsic.

1.2. Failure at the nanoscale

- **GBS:** For general high-angle boundaries the DSC lattice is commensurate only statistically.
- **Nanocrystals at $d < 10b$:** Standard plasticity predicts hardening not observed (inverse Hall–Petch).
- **Metallic glasses:** Deformation via STZs; no dislocations.

1.3. What OSET achieves

OSET replaces the lattice-dependent, singular, athermal dislocation with a geometry-defined, non-singular, thermally activated oblate spheroidal excitation. Dislocations are *derived* as the large-N collective limit. Three inputs: W (OSTZ radius), γ_0 (shear eigenstrain), G (shear modulus). All other dislocation quantities emerge.

2. The Fundamental Entity: the OSTZ

2.1. Geometric definition

An OSTZ is an oblate spheroid that has undergone a shear eigenstrain:

$$a_1 = a_2 = W, \quad a_3 = W/2, \quad \alpha \equiv a_3/a_1 = \tfrac{1}{2},$$
$$V_0 = \tfrac{4\pi}{3}a_1^2 a_3 = \tfrac{2\pi}{3}W^3. \tag{2}$$

The flat face lies in the x_1–x_2 glide plane. The eigenstrain for grain-boundary sliding (GBS):

$$\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2 \text{ (shear)}, \quad \varepsilon_{ii}^* = \varepsilon_0/3 \text{ (dilat.)} \tag{3}$$

This geometry makes S_{1313}, not S_{1212}, the Eshelby component governing stored elastic energy.

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2.2. Eshelby solution for the OSTZ

By the Eshelby inclusion theorem [?], the stress inside a homogeneous ellipsoidal inclusion is *uniform*:

$$\varepsilon_{ij}^{\text{in}} = S_{ijkl}\varepsilon_{kl}^{*}, \tag{4}$$

$$\sigma_{ij}^{\text{in}} = C_{ijkl}(S_{klmn} - I_{klmn})\varepsilon_{mn}^{*}. \tag{5}$$

Uniformity holds only for ellipsoids — the only shapes for which S_{ijkl} is spatially uniform.

The relevant Eshelby component for the oblate spheroid [?]:

$$S_{1313} = \frac{(1+\alpha^2)I_{13} + (1-2\nu)(I_1+I_3)}{16\pi(1-\nu)}, \tag{6}$$

where $\Delta(s) = \sqrt{(a_1^2+s)(a_2^2+s)(a_3^2+s)}$,

$$I_\alpha = 2\pi a_1 a_2 a_3 \int_0^\infty \frac{\mathrm{d}s}{(a_\alpha^2+s)\Delta(s)}, \tag{7}$$

$$I_{\alpha\beta} = 2\pi a_1 a_2 a_3 \int_0^\infty \frac{\mathrm{d}s}{(a_\alpha^2+s)(a_\beta^2+s)\Delta(s)},$$

with sum rule $I_1 + I_2 + I_3 = 4\pi$.

2.2.1. Derivation of I_1

Starting point.. Specialize to $a_1 = a_2 = 1$, $a_3 = \alpha$. Then $\Delta(s) = (1+s)\sqrt{\alpha^2+s}$ and

$$I_1 = 2\pi\alpha \int_0^\infty \frac{\mathrm{d}s}{(1+s)^2\sqrt{\alpha^2+s}}. \tag{8}$$

Substitution.. Let $t = \sqrt{\alpha^2+s}$, so $s = t^2 - \alpha^2$, $\mathrm{d}s = 2t\,\mathrm{d}t$, $\sqrt{\alpha^2+s} = t$, and $1+s = t^2 + (1-\alpha^2) \equiv t^2 + e^2$ (where $e^2 \equiv 1-\alpha^2$). New limits: $t : \alpha \to \infty$. Substituting:

$$I_1 = 4\pi\alpha \int_\alpha^\infty \frac{\mathrm{d}t}{(t^2+e^2)^2}. \tag{9}$$

Antiderivative.. The reduction formula (proved by differentiating the right side):

$$\int \frac{\mathrm{d}t}{(t^2+e^2)^2} = \frac{t}{2e^2(t^2+e^2)} + \frac{\arctan(t/e)}{2e^3} + C. \tag{10}$$

Proof. Let $F = t/[2e^2(t^2+e^2)]$, $G_f = \arctan(t/e)/(2e^3)$:

$$F' = \frac{1}{2e^2}\cdot\frac{(t^2+e^2) - t\cdot 2t}{(t^2+e^2)^2} = \frac{e^2-t^2}{2e^2(t^2+e^2)^2},$$

$$G_f' = \frac{1}{2e^3}\cdot\frac{1/e}{1+(t/e)^2} = \frac{1}{2e^2(t^2+e^2)}.$$

Sum: $F' + G_f' = [(e^2-t^2) + (t^2+e^2)]/[2e^2(t^2+e^2)^2] = 1/(t^2+e^2)^2$. □

Evaluating at the bounds.. *Upper bound* ($t \to \infty$): $F \to 0$, $\arctan \to \pi/2$, so $[F + G_f]_{t\to\infty} = \pi/(4e^3)$.

Lower bound ($t = \alpha$): note $\alpha^2 + e^2 = 1$, so $F|_\alpha = \alpha/(2e^2)$, $G_f|_\alpha = \arctan(\alpha/e)/(2e^3)$. Hence:

$$\int_\alpha^\infty \frac{\mathrm{d}t}{(t^2+e^2)^2} = \frac{\pi}{4e^3} - \frac{\alpha}{2e^2} - \frac{\arctan(\alpha/e)}{2e^3}$$
$$= \frac{1}{2e^3}\left[\frac{\pi}{2} - \arctan\left(\frac{\alpha}{e}\right)\right] - \frac{\alpha}{2e^2}. \tag{11}$$

Identity $\pi/2 - \arctan(\alpha/e) = \arccos\alpha$.. Set $\alpha = \cos\phi$ ($e = \sin\phi$). Then $\arctan(\cos\phi/\sin\phi) = \arctan(\cot\phi) = \pi/2 - \phi$, so $\pi/2 - \arctan(\alpha/e) = \phi = \arccos\alpha$. Substituting into Eq. (??) and multiplying by $4\pi\alpha$, and replacing $e^3 = (1-\alpha^2)^{3/2}$:

$$\boxed{I_1 = \frac{2\pi\alpha}{(1-\alpha^2)^{3/2}}\left[\arccos\alpha - \alpha\sqrt{1-\alpha^2}\right], \quad I_3 = 4\pi - 2I_1.} \tag{12}$$

2.2.2. Derivation of $I_{13} = (I_3 - I_1)/(1-\alpha^2)$

Writing the integrals explicitly ($\Delta = (1+s)\sqrt{\alpha^2+s}$):

$$I_1 = 2\pi\alpha \int_0^\infty \frac{\mathrm{d}s}{(1+s)^2\sqrt{\alpha^2+s}},$$
$$I_3 = 2\pi\alpha \int_0^\infty \frac{\mathrm{d}s}{(1+s)(\alpha^2+s)^{3/2}},$$
$$I_{13} = 2\pi\alpha \int_0^\infty \frac{\mathrm{d}s}{(1+s)^2(\alpha^2+s)^{3/2}}.$$

Forming $I_3 - I_1$ and factoring out $1/[(1+s)\sqrt{\alpha^2+s}]$:

$$I_3 - I_1 = 2\pi\alpha \int_0^\infty \frac{1}{(1+s)\sqrt{\alpha^2+s}}\left[\frac{1}{\alpha^2+s} - \frac{1}{1+s}\right]\mathrm{d}s.$$

The bracket equals $(1-\alpha^2)/[(\alpha^2+s)(1+s)]$:

$$I_3 - I_1 = (1-\alpha^2)\cdot 2\pi\alpha \int_0^\infty \frac{\mathrm{d}s}{(1+s)^2(\alpha^2+s)^{3/2}} = (1-\alpha^2)I_{13}.$$

Therefore:

$$\boxed{I_{13} = \frac{I_3 - I_1}{1-\alpha^2}.} \quad \square \tag{13}$$

2.2.3. Numerical evaluation ($\alpha = 1/2$, $\nu = 1/3$)

Step 1: compute I_1. $e = \sqrt{1-0.25} = 0.8660$. Bracket: $\arccos(0.5) - 0.5 \times 0.8660 = 1.0472 - 0.4330 = 0.6142$. $(0.75)^{3/2} = 0.6495$. $I_1 = \pi \times 0.6142/0.6495 = \mathbf{2.972}$.

Step 2: I_3 and I_{13}. $I_3 = 4\pi - 2 \times 2.972 = 12.566 - 5.944 = \mathbf{6.622}$. $I_{13} = (6.622 - 2.972)/0.75 = 3.650/0.75 = \mathbf{4.867}$.

Step 3: S_{1313} from Eq. (??). Numerator: $1.25 \times 4.867 + (1-2/3) \times 9.594 = 6.084 + 3.198 = 9.282$. Denominator: $16\pi \times (2/3) = 33.51$. $S_{1313} = 9.282/33.51 = \mathbf{0.2772}$.

Step 4: β_1. $\beta_1 = 1 - 2 \times 0.2772 = \mathbf{0.4456} \approx 0.446$.

2.2.4. Limiting-case verification

Sphere ($\alpha \to 1$).. For a sphere all semi-axes equal: $I_1 = I_2 = I_3 = 4\pi/3$. Direct integration: $I_{13}^{\text{sphere}} = 2\pi\int_0^\infty (1+s)^{-7/2}\mathrm{d}s = 2\pi[u^{-5/2}/(-5/2)]_1^\infty = 4\pi/5$. Substituting ($1+\alpha^2 \to 2$, $I_1 + I_3 = 8\pi/3$):

$$S_{1313}^{\text{sphere}} = \frac{4-5\nu}{15(1-\nu)}. \tag{14}$$

For $\nu = 1/3$: $S_{1313}^{\text{sphere}} = 7/30 = 0.233$, $\beta_1^{\text{sphere}} = 8/15 \approx 0.533$.

Thin disk ($\alpha \to 0$).. $I_3 \to 4\pi$, $I_{13} \to 4\pi$, $S^{\rm disk}_{1313} = \frac{1}{2}$, $\beta_1 \to 0$: a crack-like disk stores no shear energy. The OSTZ aspect ratio $\alpha = 1/2$ gives the correct intermediate.

2.2.5. Dilatational constraint factor β_2

For a spherical inclusion with purely dilatational eigenstrain $\varepsilon^*_{ij} = (\varepsilon_0/3)\delta_{ij}$, the sphere Eshelby tensor satisfies:

$$S^{\rm sphere}_{ijkk} = \frac{1+\nu}{3(1-\nu)}\delta_{ij}. \tag{15}$$

This follows from the known sphere values [?] $S_{1111} = (7-5\nu)/[15(1-\nu)]$, $S_{1122} = (5\nu-1)/[15(1-\nu)]$, summing: $S_{1111} + S_{1122} + S_{1133} = (1+\nu)/[3(1-\nu)]$. Full derivation of β_2 follows in §??; the result:

$$\boxed{\beta_2 = \frac{4(1+\nu)}{9(1-\nu)}.} \tag{16}$$

For $\nu = 1/3$: $\beta_2 = 4(4/3)/[9(2/3)] = 0.889$.

Two formulas for β_2 exist: (a) exact oblate formula $\beta_2^{\rm oblate} = 1 - S_{3333}(1-2\nu)/[2(1-\nu)] = 0.816$; (b) spherical dilatation approximation $\beta_2^{\rm sphere} = 0.889$ [?]. They differ by $\sim 9\%$ at $\nu = 1/3$; since $\beta_2\varepsilon_0^2 \ll \beta_1\gamma_0^2$, the difference is $< 2\%$ of ΔF_0 and inconsequential.

Table 1: Eshelby components for $\alpha = 1/2$.

ν	S_{1313}	S_{1212}	S_{3333}	β_1
0.30	0.2822	0.1769	0.7264	0.4356
1/3	0.2772	0.1739	0.7364	0.4456
0.35	0.2745	0.1723	0.7418	0.4510
0.40	0.2656	0.1670	0.7596	0.4688

2.3. Note on β_1: two physical regimes

(a) Eshelby (elastic) value: $\beta_1^{\rm Esh} = 1 - 2S_{1313} \approx$ 0.44–0.49 for $\alpha = 0.5$, $\nu = 0.28$–0.44.

(b) Padmanabhan et al. effective value: $\beta_1^{\rm eff} \approx 1$.

Physical reconciliation. OSTZs at grain boundaries are embedded in GB material with $G_{\rm GB} \approx G/2$ (GB softening [?]): $\beta_1^{\rm eff} = \beta_1^{\rm Esh} \times G/G_{\rm GB} \approx 0.45 \times 2 = 0.90 \approx 1$.

2.4. OSTZ elastic strain energy

The Eshelby energy theorem: $\Delta F_0 = -\frac{1}{2}\sigma^{\rm in}_{ij}\varepsilon^*_{ij}V_0$.

2.4.1. Part 1: Shear contribution

Eigenstrain $\varepsilon^*_{13} = \varepsilon^*_{31} = \gamma_0/2$.

Step 1 (constrained strain). By minor symmetry $S_{1313} = S_{1331}$: $\varepsilon^{\rm in}_{13} = S_{1313}(\gamma_0/2 + \gamma_0/2) = S_{1313}\gamma_0$.

Step 2 (elastic strain). $e^{\rm el}_{13} = \varepsilon^{\rm in}_{13} - \varepsilon^*_{13} = (S_{1313} - \frac{1}{2})\gamma_0$.

Step 3 (interior stress). $\sigma^{\rm in}_{13} = 2Ge^{\rm el}_{13} = 2G(S_{1313} - \frac{1}{2})\gamma_0$.

Step 4 (energy contraction). Both $(1,3)$ and $(3,1)$ contribute equally:

$$\Delta F_0^{\rm shear} = -\tfrac{1}{2}(\sigma^{\rm in}_{13}\varepsilon^*_{13} + \sigma^{\rm in}_{31}\varepsilon^*_{31})V_0 = -\tfrac{1}{2}\sigma^{\rm in}_{13}\gamma_0V_0.$$

Substituting: $\Delta F_0^{\rm shear} = -\frac{1}{2}\cdot 2G(S_{1313} - \frac{1}{2})\gamma_0 \cdot \gamma_0V_0 = G(\frac{1}{2} - S_{1313})\gamma_0^2V_0$.

$$\boxed{\Delta F_0^{\rm shear} = \tfrac{1}{2}\beta_1G\gamma_0^2V_0, \quad \beta_1 \equiv 1 - 2S_{1313}.} \tag{17}$$

Since $S_{1313} < \frac{1}{2}$ for any physical oblate spheroid, $\beta_1 > 0$ and the shear energy is positive definite.

2.4.2. Part 2: Dilatational contribution

Eigenstrain $\varepsilon^*_{ij} = (\varepsilon_0/3)\delta_{ij}$.

Step 1 (constrained strain inside a sphere). From Eqs. (??) and (??):

$$\varepsilon^{\rm in}_{ij} = S_{ijkl}\varepsilon^*_{kl} = \frac{\varepsilon_0}{3}\cdot\frac{1+\nu}{3(1-\nu)}\delta_{ij} = \frac{(1+\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}. \tag{18}$$

Step 2 (elastic strain).

$$\begin{aligned} e^{\rm el}_{ij} = \varepsilon^{\rm in}_{ij} - \varepsilon^*_{ij} &= \varepsilon_0\left[\frac{1+\nu}{9(1-\nu)} - \frac{1}{3}\right]\delta_{ij} \\ &= \varepsilon_0\cdot\frac{(1+\nu)-3(1-\nu)}{9(1-\nu)}\delta_{ij} = -\frac{2(1-2\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}. \end{aligned} \tag{19}$$

(Numerator: $1 + \nu - 3 + 3\nu = 4\nu - 2 = -2(1-2\nu)$.) Volumetric trace: $e^{\rm el}_{kk} = -6(1-2\nu)\varepsilon_0/[9(1-\nu)] = -2(1-2\nu)\varepsilon_0/[3(1-\nu)]$.

Step 3 (interior stress). Using $\lambda = 2G\nu/(1-2\nu)$:

$$\begin{aligned} \lambda e^{\rm el}_{kk} &= \frac{2G\nu}{1-2\nu}\cdot\frac{-2(1-2\nu)\varepsilon_0}{3(1-\nu)} = -\frac{4G\nu\varepsilon_0}{3(1-\nu)}, \\ 2Ge^{\rm el}_{11} &= -\frac{4G(1-2\nu)\varepsilon_0}{9(1-\nu)}. \end{aligned}$$

Adding with common denominator $9(1-\nu)$:

$$\begin{aligned} \sigma^{\rm in}_{11} &= -\frac{12G\nu\varepsilon_0}{9(1-\nu)} - \frac{4G(1-2\nu)\varepsilon_0}{9(1-\nu)} \\ &= -\frac{G\varepsilon_0}{9(1-\nu)}[12\nu + 4 - 8\nu] = -\frac{4G(1+\nu)\varepsilon_0}{9(1-\nu)}. \end{aligned} \tag{20}$$

Step 4 (energy contraction). Three diagonal components contribute equally: $\sigma^{\rm in}_{ij}\varepsilon^*_{ij} = 3\sigma^{\rm in}_{11}(\varepsilon_0/3) = \sigma^{\rm in}_{11}\varepsilon_0$.

$$\boxed{\Delta F_0^{\rm dilat} = \tfrac{1}{2}\beta_2G\varepsilon_0^2V_0, \quad \beta_2 = \frac{4(1+\nu)}{9(1-\nu)}.} \tag{21}$$

2.4.3. Total activation energy

Shear and dilatational eigenstrains involve orthogonal tensor components (ε^*_{13} vs. ε^*_{ii}); their cross-terms vanish upon contraction with the isotropic stiffness tensor. Therefore:

$$\boxed{\Delta F_0 = \tfrac{1}{2}(\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2)GV_0, \quad V_0 = \tfrac{2\pi}{3}W^3.} \tag{22}$$

Numerical check: $\Delta F_0 = \frac{1}{2}(0.00446 + 0.00222) \times 2.2 \times 10^{10} \times 0.884 \times 10^{-27} = 6.5 \times 10^{-20}$ J$= 0.406$ eV≈ 0.38 eV.

Table 2: Canonical OSTZ parameters.

Parameter	Symbol	GB/superpl.	Crystal
OSTZ radius	W	$2.5b$	$\approx b$
Shear eigenstrain	γ_0	0.10	0.12
Dilat. eigenstrain	ε_0	0.05	—
Shear constraint	β_1	0.446	≈ 1
Dilat. constraint	β_2	0.889	—
Activation energy	ΔF_0	0.38 eV	—
Critical number	N_c	4	≈ 8

3. Far-Field Stress: The Dipole Structure

3.1. Why a dipole, not a monopole

The force monopole vanishes:

$$F_k = \oint_{\partial V_0} \sigma^*_{ij} n_j \, \mathrm{d}S = C_{ijkl}\varepsilon^*_{kl} \underbrace{\oint_{\partial V_0} n_j \, \mathrm{d}S}_{=0} = 0. \tag{23}$$

(The surface integral of the outward normal over any closed surface vanishes by the divergence theorem.) The leading far-field term is therefore the *force-dipole* at order r^{-3}.

The full multipole expansion [?] is

$$\sigma_{ij}(\mathbf{r}) = \underbrace{F_k \mathcal{G}_{ijk}}_{\text{monopole } r^{-2}} + \underbrace{M_{kl}\partial_l \mathcal{G}_{ijk}}_{\text{dipole } r^{-3}} + \underbrace{Q_{klm}\partial_l\partial_m \mathcal{G}_{ijk}}_{r^{-4}} + \cdots \tag{24}$$

Since $F_k = 0$, the leading nonzero term is the dipole. The moment tensor [?] is

$$M_{kl} = C_{ijkl}\varepsilon^*_{ij} V_0 \;\Rightarrow\; M_{13} = M_{31} = G\gamma_0 V_0, \tag{25}$$

describing a double couple: two force pairs of equal and opposite sign, separated by the OSTZ diameter $2W$.

3.2. Step A: Kelvin Green's function and its derivative

The elastic displacement due to a unit point force at the origin in an infinite isotropic medium (Kelvin–Somigliana solution [?]):

$$\mathcal{G}_{ij}(\mathbf{r}) = A\left[\frac{(3-4\nu)\delta_{ij}}{r} + \frac{x_i x_j}{r^3}\right], \quad A = \frac{1}{16\pi G(1-\nu)}. \tag{26}$$

Differentiating the first term.. Since $\partial_l(1/r) = -x_l/r^3$:

$$\partial_l\left[\frac{(3-4\nu)\delta_{ij}}{r}\right] = -\frac{(3-4\nu)\delta_{ij}x_l}{r^3}.$$

Differentiating the second term (product rule).. $\partial x_i/\partial x_l = \delta_{il}$, $\partial(r^{-3})/\partial x_l = -3x_l/r^5$:

$$\partial_l\left(\frac{x_i x_j}{r^3}\right) = \frac{\delta_{il}x_j + \delta_{jl}x_i}{r^3} - \frac{3x_l x_i x_j}{r^5}.$$

Combined result..

$$\frac{\partial \mathcal{G}_{ij}}{\partial x_l} = A\Big[-\frac{(3-4\nu)\delta_{ij}x_l}{r^3} + \frac{\delta_{il}x_j + \delta_{jl}x_i}{r^3} - \frac{3x_l x_i x_j}{r^5}\Big]. \tag{27}$$

3.3. Step B: Displacement from the moment tensor

For an Eshelby eigenstrain inclusion, the far-field exterior displacement is [? ?] $u_i = -M_{kl}\partial_l \mathcal{G}_{ik}$. With $M_{13} = M_{31} = G\gamma_0 V_0$ (all other components zero), the only nonzero contributions are $(k,l) = (1,3)$ and $(3,1)$.

Substituting Eq. (??) (with $\delta_{13} = 0$):

$$\frac{\partial \mathcal{G}_{i1}}{\partial x_3} = A\left[-\frac{(3-4\nu)\delta_{i1}x_3}{r^3} + \frac{\delta_{i3}x_1}{r^3} - \frac{3x_3 x_i x_1}{r^5}\right],$$
$$\frac{\partial \mathcal{G}_{i3}}{\partial x_1} = A\left[-\frac{(3-4\nu)\delta_{i3}x_1}{r^3} + \frac{\delta_{i1}x_3}{r^3} - \frac{3x_1 x_i x_3}{r^5}\right].$$

Adding term by term, the $\delta_{i1}x_3$ coefficient is $1-(3-4\nu) = 4\nu - 2 = 2(2\nu-1)$ (same for $\delta_{i3}x_1$), and the r^{-5} terms add to give $-6Ax_1x_3x_i/r^5$:

$$u_i = -G\gamma_0 V_0 A\Big[\frac{2(2\nu-1)(\delta_{i1}x_3 + \delta_{i3}x_1)}{r^3} - \frac{6x_1x_3x_i}{r^5}\Big]. \tag{28}$$

Components relevant to σ_{13}:

$$u_1 = -G\gamma_0 V_0 A\left[\frac{2(2\nu-1)x_3}{r^3} - \frac{6x_1^2 x_3}{r^5}\right], \tag{29}$$

$$u_3 = -G\gamma_0 V_0 A\left[\frac{2(2\nu-1)x_1}{r^3} - \frac{6x_1 x_3^2}{r^5}\right]. \tag{30}$$

3.4. Step C: Stress from displacement gradients

For the shear component ($\delta_{13} = 0$, so the λ-term in Hooke's law vanishes): $\sigma_{13} = G(\partial_3 u_1 + \partial_1 u_3)$.

Auxiliary results.. $\partial_3(x_3/r^3) = (r^2 - 3x_3^2)/r^5$; $\partial_3(x_3/r^5) = (r^2 - 5x_3^2)/r^7$; likewise for x_1.

Computing $\partial_3 u_1$.. Differentiating Eq. (??) with respect to x_3:

$$\partial_3 u_1 = -G\gamma_0 V_0 A\Big[\frac{2(2\nu-1)(r^2 - 3x_3^2)}{r^5} - \frac{6x_1^2(r^2 - 5x_3^2)}{r^7}\Big]. \tag{31}$$

Computing $\partial_1 u_3$.. By the same procedure (swap $x_1 \leftrightarrow x_3$):

$$\partial_1 u_3 = -G\gamma_0 V_0 A\Big[\frac{2(2\nu-1)(r^2 - 3x_1^2)}{r^5} - \frac{6x_3^2(r^2 - 5x_1^2)}{r^7}\Big]. \tag{32}$$

Summing and converting to unit vectors $\hat{r}_i = x_i/r$.. $1/r^5$ group: $2(2\nu-1)[2-3(\hat{r}_1^2+\hat{r}_3^2)]/r^3$. $1/r^7$ group, expanding $(x_1^2+x_3^2)r^2-10x_1^2x_3^2$: $[-6(\hat{r}_1^2+\hat{r}_3^2)+60\hat{r}_1^2\hat{r}_3^2]/r^3$. Collecting the $(\hat{r}_1^2+\hat{r}_3^2)$ terms: $-6(2\nu-1)-6=-12\nu$ (verify: $12\nu-6+6=12\nu$). Hence:

$$\partial_3 u_1 + \partial_1 u_3 = -G\gamma_0 V_0 A \cdot \frac{1}{r^3} \times \left[4(2\nu-1) - 12\nu(\hat{r}_1^2+\hat{r}_3^2) + 60\hat{r}_1^2\hat{r}_3^2\right]. \quad (33)$$

3.5. Step D: Angular tensor and far-field formula

Multiplying Eq. (**??**) by G and inserting $A = 1/[16\pi G(1-\nu)]$:

$$\sigma_{13} = \frac{G\gamma_0 V_0}{16\pi(1-\nu)r^3}\left[-4(2\nu-1)+12\nu(\hat{r}_1^2+\hat{r}_3^2)-60\hat{r}_1^2\hat{r}_3^2\right].$$

Writing this as $G\gamma_0 V_0 \mathcal{T}_{13}/[2\pi(1-\nu)r^3]$ requires dividing the bracket by 8 (ratio of prefactors $2\pi/16\pi = 1/8$):

$$\boxed{\mathcal{T}_{13}(\hat{\mathbf{r}}) = \tfrac{1}{2} - \nu + \tfrac{3\nu}{2}(\hat{r}_1^2+\hat{r}_3^2) - \tfrac{15}{2}\hat{r}_1^2\hat{r}_3^2.} \quad (34)$$

$$\boxed{\sigma_{13}(\mathbf{r}) = \frac{G\gamma_0 V_0}{2\pi(1-\nu)} \cdot \frac{\mathcal{T}_{13}(\hat{\mathbf{r}})}{r^3}.} \quad (35)$$

The r^{-3} decay follows from differentiating the Kelvin r^{-1} Green's function twice: two orders faster than the Volterra r^{-1} field.

3.6. Step E: Verification in special directions

Glide plane ($x_3 = 0$, $\hat{r}_3 = 0$).. $\mathcal{T}_{13}|_{\hat{r}_3=0} = \frac{1}{2} - \nu + (3\nu/2)\hat{r}_1^2$. Expanding $r^2 = x_1^2 + x_2^2$ and collecting:

$$\sigma_{13}(x_1, x_2, 0) = \frac{G\gamma_0 V_0}{4\pi(1-\nu)r^5}[(1+\nu)x_1^2 + (1-2\nu)x_2^2]. \quad (36)$$

Positive definite for $0 < \nu < \frac{1}{2}$. ✓

On-axis ($\hat{r}_1 = 0$, $\hat{r}_3 = 1$).. $\mathcal{T}_{13} = \frac{1}{2} - \nu + 3\nu/2 = (1+\nu)/2 > 0$. By oblate axial symmetry, the same result holds along $\hat{x}_1$. ✓

Nodal surface.. Setting $\mathcal{T}_{13} = 0$ yields no real solution for $0 < \nu < \frac{1}{2}$: there is no nodal surface in the half-space $x_3 \geq 0$. The OSTZ promotes shear in all glide-plane directions (cooperative nucleus).

3.7. Glide-plane stress: six-step derivation

Working at $x_3 = 0$ ($r^2 = x^2 + y^2$, $x \equiv x_1$, $y \equiv x_2$).

Step 1 (off-diagonal $\mathcal{G}_{13}$). $\partial^2(\mathcal{G}_{13})/(\partial x_1 \partial x_3)$ at $z = 0$: gives $(y^2 - 2x^2)/r^5$ (using $\partial_3(x_1 x_3/r^3)|_{z=0} = x_1/r^3$, then ∂_1).

Step 2 (diagonal $\mathcal{G}_{33}$). At $z = 0$: $\mathcal{G}_{33}|_{z=0} = A(3-4\nu)/r$. $\partial_1^2(1/r) = (3x^2 - r^2)/r^5 = (2x^2 - y^2)/r^5$:

$$\partial_1^2 \mathcal{G}_{33}\big|_{z=0} = A(3-4\nu)(2x^2 - y^2)/r^5.$$

Step 3 (diagonal $\mathcal{G}_{11}$). At $z = 0$: $\mathcal{G}_{11}|_{z=0} = A[(3-4\nu)/r + x^2/r^3]$. $\partial_3^2(1/r)|_{z=0} = -1/r^3$; $\partial_3^2(1/r^3)|_{z=0} = -3/r^5$:

$$\partial_3^2 \mathcal{G}_{11}\big|_{z=0} = A[-(3-4\nu)/r^3 - 3x^2/r^5].$$

Expanding: $-(3-4\nu)r^2 - 3x^2 = -(6-4\nu)x^2 - (3-4\nu)y^2$.

Step 4 (collect numerator, units of A/r^5).

$$\begin{aligned} x^2 \text{ coeff.: } & 2(y^2 - 2x^2) \to -4 \\ & + (6-8\nu) - (3-4\nu) - 3 = -4(1+\nu), \\ y^2 \text{ coeff.: } & 2 - (3-4\nu) - (3-4\nu) = -4(1-2\nu). \end{aligned}$$

Numerator: $-4A[(1+\nu)x^2 + (1-2\nu)y^2]/r^5$.

Step 5 (assemble).

$$\sigma_{13}(x, y, 0) = \frac{G\gamma_0 V_0[(1+\nu)x^2 + (1-2\nu)y^2]}{4\pi(1-\nu)r^5}.$$

Step 6 (re-express with b_{eff}, restore finite W). $G\gamma_0 V_0 = G(\gamma_0)\frac{2\pi W^3}{3} = Gb_{\text{eff}}\frac{2\pi W^2}{3}$ (using $b_{\text{eff}} = \gamma_0 W$). Then $G\gamma_0 V_0/[4\pi(1-\nu)] = Gb_{\text{eff}} W^2/[6(1-\nu)]$. Regularising $r^2 \to r^2 + W^2$:

$$\boxed{\sigma_{13}(x, y, 0) = \frac{Gb_{\text{eff}} W^2[(1+\nu)x^2 + (1-2\nu)y^2]}{6(1-\nu)(x^2 + y^2 + W^2)^{5/2}}.} \quad (37)$$

Positive definite; finite at the origin.

Physical interpretation.. Along $\hat{x}$ ($y = 0$): promotes forward shear. Along $\hat{y}$ ($x = 0$): promotes lateral shear. No nodal planes in the glide plane: the OSTZ is a *cooperative nucleus.*

3.8. Effective Burgers vector

Method 1 (kinematic, exact): $b_{\text{eff}} = \gamma_0 \times 2a_3 = \gamma_0 W$.

Method 2 (moment tensor): $M_{13} = G\gamma_0 V_0 = Gb_{\text{eff}}\pi W^2 \Rightarrow b_{\text{eff}} = 2\gamma_0 W/3$ (underestimates 33%).

Method 3 (Lorentzian): b_{eff} fixed by normalisation; not an independent check.

Primary definition: $\boxed{b_{\text{eff}} = \gamma_0 W}$.

Table 3: Single OSTZ vs. Volterra dislocation.

Property	Single OSTZ	Volterra
Stress decay	r^{-3} (dipole)	r^{-1} (monopole)
Topo. charge	0 (neutral)	$\pm b$
Net b	0	b
Core stress	Finite ($\leq G\gamma_0$)	$\to \infty$
Influence	$\sim W$	∞
Elastic E	$\frac{1}{2}\beta_1\gamma_0^2 GV_0$	$Gb^2 L\ln(R/r_0)$
Thermal act.	Intrinsic	Appended

4. Emergence of Dislocations from OSTZ Chains

4.1. Lorentzian dislocation density: four steps

4.1.1. Step A: 2D dislocation density (BCS compliance)

The Bilby–Cottrell–Swinden (BCS) compliance [?] $\delta u_1 = 2(1-\nu)\sigma_{13}/G$ relates local slip to traction. The shear stress in the isotropic approximation:

$$\sigma_{13}(x,y,0) \approx \frac{G b_{\text{eff}} W^2}{2\pi(1-\nu)(x^2+y^2+W^2)^{3/2}}. \tag{38}$$

(This replaces the exact angular form by a spherically symmetric approximation, valid because the true field $\sim r^{-3}$ converges absolutely and is positive everywhere in the glide plane.) Carrying out the cancellations (G with $1/G$; $2(1-\nu)$ with $2\pi(1-\nu)$ leaving $1/\pi$):

$$\rho_{2D}(x,y) = \frac{b_{\text{eff}} W^2}{\pi(x^2+y^2+W^2)^{3/2}}. \tag{39}$$

Normalisation check. Converting to polar coordinates $r^2 = x^2+y^2$:

$$\iint \rho_{2D}\,\mathrm{d}x\,\mathrm{d}y = \frac{b_{\text{eff}} W^2}{\pi} \cdot 2\pi \int_0^\infty \frac{r\,\mathrm{d}r}{(r^2+W^2)^{3/2}}.$$

Substitution $u = r^2+W^2$, $du = 2r\,dr$: $\int = \frac{1}{2}[-2/\sqrt{u}]_{W^2}^{\infty} = 1/W$. Therefore $\iint \rho_{2D} = 2b_{\text{eff}}W$. (The result is $2b_{\text{eff}}W$, not b_{eff}, because ρ_{2D} is slip per unit glide-plane area; the Abel projection recovers the correct normalisation below.)

4.1.2. Step B: Abel projection

Integrate over y with $a^2 = x^2+W^2$; let $y = a\tan\theta$:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}y}{(a^2+y^2)^{3/2}} = \frac{1}{a^2}\int_{-\pi/2}^{\pi/2} \cos\theta\,\mathrm{d}\theta = \frac{2}{x^2+W^2}. \tag{40}$$

Raw 1D density: $\rho_1^{\text{raw}} = 2b_{\text{eff}}W^2/[\pi(x^2+W^2)]$, which integrates to $2b_{\text{eff}}W$.

4.1.3. Step C: Lorentzian (normalised)

Divide by $2W$ to enforce $\int \rho_1\,\mathrm{d}x = b_{\text{eff}}$:

$$\boxed{\rho_1(x) = \frac{b_{\text{eff}}}{\pi} \cdot \frac{W}{x^2+W^2}.} \tag{41}$$

Check: $\int_{-\infty}^{\infty} \rho_1\,\mathrm{d}x = (b_{\text{eff}}W/\pi)(\pi/W) = b_{\text{eff}}$. □

The Lorentzian is the unique non-negative distribution integrating to b_{eff} with width W and reducing to a Dirac delta as $W \to 0$. Both OSET and the P–N model are governed by the same Cauchy–Hilbert Green's operator of planar elasticity.

4.1.4. Step D: Fourier transform

For $k > 0$, close the contour in the lower half-plane (where $e^{-ikx} \to 0$). The only enclosed pole is at $x = -iW$:

$$\text{Res}\left[\frac{We^{-ikx}}{\pi(x^2+W^2)}, -iW\right] = \frac{e^{-kW}}{-2iW}.$$

Clockwise contour ($-2\pi i \times \text{Res}$):

$$\hat{\rho}(k) = b_{\text{eff}}\, e^{-|k|W}. \tag{42}$$

At $k = 2\pi/b$: $\hat{\rho} = b_{\text{eff}}e^{-2\pi W/b}$ (Peierls lattice-filter factor). The OSTZ core acts as a low-pass filter: high-spatial-frequency lattice components are exponentially suppressed.

4.2. Theorem: the N-OSTZ chain is a Peierls–Nabarro dislocation

For N co-planar OSTZs: $\rho_N(x) = N \cdot (b_{\text{eff}}/\pi)W/(x^2+W^2)$.

The cumulative slip from $-\infty$ to x:

$$\begin{aligned} U(x) &= \int_{-\infty}^{x} \rho_N(x')\,\mathrm{d}x' \\ &= \frac{Nb_{\text{eff}}W}{\pi} \cdot \frac{1}{W}\left[\arctan\frac{x'}{W}\right]_{-\infty}^{x} \\ &= \frac{Nb_{\text{eff}}}{2} + \frac{Nb_{\text{eff}}}{\pi}\arctan\frac{x}{W}. \end{aligned} \tag{43}$$

(Using $\arctan(-\infty) = -\pi/2$.)

Setting $b \equiv Nb_{\text{eff}}$ and $\zeta \equiv W$:

$$U(x) = \frac{b}{2} + \frac{b}{\pi}\arctan\frac{x}{\zeta}. \tag{44}$$

This is exactly the Peierls–Nabarro profile [? ?]. *Limits:* $U(-\infty) = 0$, $U(+\infty) = b$, $U(0) = b/2$. □

The continuum dislocation density is $\rho(x) = \partial U/\partial x$:

$$\rho(x) = \frac{Nb_{\text{eff}}}{\pi} \cdot \frac{W}{x^2+W^2} = \rho_N(x). \quad \square$$

Corollary. The core half-width $\zeta = W$ is derived from OSTZ geometry, not a free parameter. The Peierls–Nabarro model is not an independent theory: it is the continuum limit of the OSTZ ensemble.

4.3. Chain stress field and Volterra limit

$$\sigma_{13}^{(N)}(x) = \frac{G}{2\pi(1-\nu)}\,\text{P.V.}\int_{-\infty}^{\infty} \frac{\rho_N(x')}{x-x'}\,\mathrm{d}x'. \tag{45}$$

Partial fractions.. Decompose $W/[(x'^2+W^2)(x-x')]$: $W = A(x'^2+W^2) + (Bx'+C)(x-x')$. Matching powers of x': x'^2: $A = B$; x'^1: $Bx - C = 0 \Rightarrow C = Ax$; constant: $A(x^2+W^2) = W \Rightarrow A = W/(x^2+W^2)$. Hence:

$$\frac{W}{(x'^2+W^2)(x-x')} = \frac{W}{x^2+W^2}\left[\frac{1}{x-x'} + \frac{x'+x}{x'^2+W^2}\right].$$

Term 1.. P.V. $\int \mathrm{d}x'/(x-x') = 0$ (integrand is odd about $x' = x$).

Term 2.. $\int x'/(x'^2+W^2)\,\mathrm{d}x' = 0$ (odd in x'). $\int \mathrm{d}x'/(x'^2+W^2) = \pi/W$ (via $x' = W\tan\theta$). So Term 2 gives $\pi x/(x^2+W^2)$.

Combining:.

$$\boxed{\sigma_{13}^{(N)}(x) = \frac{GNb_{\mathrm{eff}}}{2\pi(1-\nu)} \cdot \frac{x}{x^2+W^2}.} \tag{46}$$

Non-singular at $x = 0$; maximum at $x = W$.

Volterra limit ($W \to 0$).. $\lim_{W\to 0} x/(x^2+W^2) = 1/x$, so:

$$\sigma_{13}^{\mathrm{Volterra}} = \frac{Gb}{2\pi(1-\nu)x}. \tag{47}$$

The Volterra singularity is a collective limit; no single OSTZ is singular.

Burgers circuit.. $\oint_C \mathrm{d}u_1 = Nb_{\mathrm{eff}} = b$. Quantisation $b \in \Lambda_{\mathrm{Bravais}}$ arises when the lattice constrains N to N_c.

4.4. Critical cluster number

Setting $Nb_{\mathrm{eff}} = b_{\mathrm{latt}}$:

$$\boxed{N_c = \frac{b_{\mathrm{latt}}}{\gamma_0 W}.} \tag{48}$$

FCC Cu ($W = b$, $\gamma_0 = 0.12$): $N_c \approx 8$. GB regime ($W = 2.5b$, $\gamma_0 = 0.10$): $N_c = 4$ (confirmed experimentally in Zn–22Al [?]).

Why dislocations do not form in amorphous materials: no periodic potential to lock displacements at quantised values $b_i \in \Lambda_{\mathrm{Bravais}}$, so OSTZs activate individually or in small clusters ($N \ll N_c$): STZ-like deformation.

5. Dislocation Quantities as Derived Results

5.1. Theoretical shear strength

The cascade instability threshold: $\tau^*\gamma_0 V_0 = \Delta F_0 = \frac{1}{2}\beta_1\gamma_0^2 G V_0$:

$$\boxed{\tau^* = \frac{\beta_1\gamma_0 G}{2}.} \tag{49}$$

For $\beta_1 \approx 1$, $\gamma_0 = 0.12$: $\tau^* \approx G/17$ (Frenkel range $G/10$–$G/30$).

5.2. Peierls stress

5.2.1. OSET-derived amplitude

Setting $\tau^* = (2\pi/b)V_0^{\mathrm{OSET}}$:

$$V_0^{\mathrm{OSET}} = \frac{\beta_1\gamma_0 Gb}{4\pi}. \tag{50}$$

For Cu: $\tau^* \approx G/17$ vs. the Frenkel estimate $G/[2(1-\nu)] \approx 0.75G$.

5.2.2. Fourier convolution

Misfit energy $E_{\mathrm{mis}} = \int V(x-x_0)\rho(x)\,\mathrm{d}x$, $V = V_0[1-\cos(2\pi x/b)]$. Using $\hat{\rho}(k_1) = be^{-2\pi W/b}$ at $k_1 = 2\pi/b$ and Euler's formula $\cos = \mathrm{Re}[e^{i\theta}]$:

$$E_{\mathrm{mis}}(x_0)\big|_{\mathrm{osc}} = -\frac{Gb^2}{4\pi(1-\nu)}\,e^{-2\pi W/b}\cos\left(\frac{2\pi x_0}{b}\right). \tag{51}$$

(Constant $V_0 b$ term dropped as it plays no role in τ_P.)

5.2.3. Peierls barrier and half-space correction

Barrier: $\Delta E_P = 2Ae^{-2\pi W/b}$ with $A = Gb^2/[4\pi(1-\nu)]$. Maximum slope: $\tau_P = (2\pi A/b^2)e^{-2\pi W/b} = Ge^{-2\pi W/b}/[2(1-\nu)]$. With half-space correction [?] ($W \to W/(1-\nu)$, prefactor $\times 2$):

$$\boxed{\tau_P = \frac{2G}{1-\nu}\exp\left(-\frac{2\pi W}{b(1-\nu)}\right).} \tag{52}$$

Cu ($W = b$, $\nu = 1/3$): $\tau_P \approx 2.1 \times 10^{-4}G$ (experiment: $\sim 10^{-4}G$).

5.3. Core energy

$E_{\mathrm{core}} = N_c\Delta F_0$. Substituting $N_c = b/(\gamma_0 W)$ and $\Delta F_0 = \pi\beta_1\gamma_0^2 GW^3/3$:

$$E_{\mathrm{core}} = \frac{b}{\gamma_0 W} \cdot \frac{\pi\beta_1\gamma_0^2 GW^3}{3} = \frac{\pi\beta_1 b\gamma_0^2 GW^3}{3\gamma_0 W}. \tag{53}$$

$$\boxed{E_{\mathrm{core}} = \frac{\pi\beta_1 b\gamma_0 GW^2}{3}.} \tag{54}$$

Writing $E_{\mathrm{core}} = \alpha Gb^2$: for Cu ($W \approx b$), $\alpha = \pi\beta_1\gamma_0/3 \approx 0.13$ (exp/DFT: 0.1–0.2).

5.4. Frank–Read critical stress

Following Frank & Read [?]:

Step 1 (line tension). $T = Gb^2/[4\pi(1-\nu)]$. Semicircle of radius $R = L/2$ has arc length $\pi L/2$:

$$E_{\mathrm{bow}} = T \cdot \pi R = \frac{Gb^2 L}{8(1-\nu)}. \tag{55}$$

Step 2 (external work). Enclosed area $\pi R^2/2 = \pi L^2/8$:

$$W_{\mathrm{ext}} = \tau_{\mathrm{FR}} \cdot b \cdot \frac{\pi L^2}{8}. \tag{56}$$

Step 3 ($W_{\mathrm{ext}} = E_{\mathrm{bow}}$). Dividing both sides by $bL/8$: $\tau_{\mathrm{FR}} \cdot \pi L = Gb/(1-\nu)$:

$$\boxed{\tau_{\mathrm{FR}} = \frac{Gb}{\pi(1-\nu)L} \approx \frac{Gb}{2L}.} \tag{57}$$

Cu at $L = 1$ μm: 14 MPa vs. classical 12.4 MPa (13%).

5.5. Stacking-fault energy

5.5.1. FCC Burgers ratios

$|b_{\text{full}}| = a_0/\sqrt{2}$; $|b_{\text{partial}}| = a_0/\sqrt{6}$:

$$\frac{|b_{\text{partial}}|}{|b_{\text{full}}|} = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}. \tag{58}$$

5.5.2. OSTZ cluster for the Shockley partial

Setting $N_p\gamma_0 W = |b_{\text{partial}}| = N_c\gamma_0 W/\sqrt{3}$: $N_p = N_c/\sqrt{3}$. Evaluating $(N_c - N_p)/N_p$:

$$N_c - N_p = N_c\left(1 - \frac{1}{\sqrt{3}}\right), \quad \frac{N_c - N_p}{N_p} = \sqrt{3}\left(1 - \frac{1}{\sqrt{3}}\right) = \sqrt{3} - 1.$$

5.5.3. SFE derivation

The energy per fault area (πW^2 per OSTZ site, N_p sites):

$$\gamma_{\text{SF}} = \frac{(N_c - N_p)\Delta F_0}{\pi W^2 N_p} = \frac{(\sqrt{3}-1)\Delta F_0}{\pi W^2}.$$

Substituting $\Delta F_0 = \pi\beta_1\gamma_0^2 G W^3/3$ and cancelling $W^3/W^2 = W$:

$$\boxed{\gamma_{\text{SF}} = \frac{(\sqrt{3}-1)\beta_1\gamma_0^2 G W}{3}.} \tag{59}$$

Cu ($G = 48.3$ GPa, $W = b = 0.2556$ nm, $\gamma_0 = 0.12$, $\beta_1 = 1$): $\gamma_{\text{SF}} = 43$ mJ/m^2 vs. experiment 45 mJ/m^2 (5%). Equation (**??**) identifies $W \approx b$ as the reason for $\gamma_{\text{SF}} \propto Gb$ across FCC metals.

Table 4: SFE predictions ($\gamma_0 = 0.12$, $W = b$, $\beta_1^{\text{eff}} = 1$).

Metal	G (GPa)	b (nm)	$\gamma_{\text{SF}}^{\text{OSET}}$ (mJ/m^2)	Exp. (mJ/m^2)
Cu	48.3	0.256	43	45
Al	26.2	0.286	26	166
Ni	76.0	0.249	67	125
Ag	30.3	0.289	31	16
Au	27.0	0.288	28	32

6. Grain-Boundary Energy

6.1. Coherent boundaries (quadratic in eigenstrain)

A coherent twin boundary (CTB) carries a coherency strain over an interfacial area. Its energy is *quadratic* in the eigenstrain — identical to the OSET SFE (**??**): $\gamma_{\text{CTB}} = (\sqrt{3}-1)\beta_1\gamma_0^2 GW/3$.

6.2. Incoherent boundaries: the Read–Shockley law

A low/high-angle boundary is a dislocation array spaced $D = b/\theta$. In OSET each dislocation is an N_c-OSTZ chain; summing the elastic dipole fields reproduces the classical dislocation line energy $E_d = (Gb^2/4\pi(1-\nu))\ln(R/r_0)$. Dividing by D with $R \sim D/2$ ($\ln(R/r_0) \to A - \ln\theta$) yields the Read–Shockley law [**?**]:

$$\gamma_{\text{GB}}(\theta) = E_0\,\theta(A - \ln\theta), \tag{60}$$
$$E_0 = \frac{Gb}{4\pi(1-\nu)}, \quad A = 1 + \ln\theta_m.$$

The GB energy scales linearly with b (dislocation array = linear in eigenstrain via $b = N_c\gamma_0 W$), not quadratically with γ_0 (activation barrier). Conflating these objects causes 5–40× error.

7. OSTZ Hamiltonian and Statistical Mechanics

7.1. Construction of the Hamiltonian

The grain boundary is modelled as a 2D array of sites i, each with binary occupation $n_i \in \{0, 1\}$.

Term 1: Self-energy..

$$\mathcal{H}_{\text{self}} = \sum_i \Delta F_{0i}\, n_i. \tag{61}$$

Term 2: Mechanical work.. Applied stress τ does work $(\tau - \tau_0)\gamma_0 V_0$ per activated site, lowering the barrier ($\tau_0 \propto L^{-1/2}$ is the threshold back-stress):

$$\mathcal{H}_{\text{mech}} = -\sum_i (\tau - \tau_0)\gamma_0 V_0\, n_i. \tag{62}$$

Term 3: Elastic interaction.. Two activated OSTZs at $\mathbf{x}_i$ and $\mathbf{x}_j$ interact via their overlapping r^{-3} fields:

$$E_{ij}^{\text{int}} = -\sigma_{13}^{(i)}(\mathbf{x}_j)\gamma_0 V_0 = -\frac{G\gamma_0^2 V_0^2 \mathcal{T}_{13}(\hat{\mathbf{r}}_{ij})}{2\pi(1-\nu)r_{ij}^3}. \tag{63}$$

The leading (dipole–dipole) term defines the interaction kernel:

$$J_{ij} = G\gamma_0^2 V_0^2\, \mathcal{K}\left(\frac{r_{ij}}{W}\right), \tag{64}$$
$$\mathcal{K}(\rho) = \frac{3\cos^2\theta_{ij} - 1}{4\pi(1-\nu)\rho^3}.$$

$J_{ij} > 0$ (cooperative) for OSTZs aligned along the slip direction ($\theta_{ij} = 0$); $J_{ij} < 0$ (anti-cooperative) perpendicular. The form is identical to the magnetic dipole–dipole interaction: a direct consequence of the shared r^{-3} Green's function structure.

$$\mathcal{H}_{\text{int}} = -\tfrac{1}{2}\sum_{i\neq j} J_{ij}\, n_i n_j. \tag{65}$$

Full Hamiltonian..

$$\mathcal{H} = \sum_i [\Delta F_{0i} - (\tau - \tau_0)\gamma_0 V_0]\, n_i - \tfrac{1}{2}\sum_{i\neq j} J_{ij}\, n_i n_j. \tag{66}$$

This is formally a *random-field Ising model* with quenched disorder ΔF_{0i} and ferromagnetic exchange J_{ij}.

7.2. Mean-field decoupling

The linearisation step.. Write $n_i = \bar{n} + \delta n_i$. Expand $n_i n_j$ and drop $\delta n_i \delta n_j$ (Bragg–Williams approximation [?]):

$$n_i n_j \approx n_i \bar{n} + n_j \bar{n} - \bar{n}^2. \tag{67}$$

Effect on the interaction term.. Substituting into $-\frac{1}{2}\sum_{i\neq j} J_{ij} n_i n_j$; the first two terms are equal by relabelling ($i \leftrightarrow j$); with translation invariance $\sum_{j\neq i} J_{ij} = zJ_0$:

$$-\tfrac{1}{2}\sum_{i\neq j} J_{ij} n_i n_j \approx -z\bar{n}J_0 \sum_i n_i + \text{const.} \tag{68}$$

Mean-field Hamiltonian.. The total factorises into independent single-site problems:

$$\begin{aligned}\mathcal{H}^{\text{MF}} &= \sum_i h^{\text{eff}} n_i + \text{const}, \\ h^{\text{eff}} &= \Delta F_0 - (\tau - \tau_0)\gamma_0 V_0 - z\bar{n}J_0.\end{aligned} \tag{69}$$

The effective barrier h^{eff} is the bare cost ΔF_0 reduced by the mechanical driving force and the cooperative elastic interaction from already-activated neighbours.

7.2.1. Self-consistency equation

Single-site partition function $\mathcal{Z}_i = 1 + e^{-h^{\text{eff}}/kT}$. Mean occupation:

$$\bar{n} = \langle n_i \rangle = \frac{e^{-h^{\text{eff}}/kT}}{1 + e^{-h^{\text{eff}}/kT}}. \tag{70}$$

Substituting h^{eff} explicitly:

$$\boxed{\bar{n} = \frac{\exp\left[-\frac{\Delta F_0 - (\tau-\tau_0)\gamma_0 V_0 - z\bar{n}J_0}{kT}\right]}{1 + \exp\left[-\frac{\Delta F_0 - (\tau-\tau_0)\gamma_0 V_0 - z\bar{n}J_0}{kT}\right]}} \tag{71}$$

This nonlinear equation is solved iteratively for $\bar{n}$.

In the *dilute limit* ($h^{\text{eff}} \gg kT$, $z\bar{n}J_0 \to 0$), $\bar{n} \approx \exp(-\Delta F_0/kT) \cdot \exp((\tau - \tau_0)\gamma_0 V_0/kT)$ — only the forward activation. The full forward–backward balance is captured by the Eyring kinetic route (§??).

7.3. Sinh flow law: step-by-step derivation

Step 1 (strain per activation).. Each OSTZ activation shifts the material by $b_{\text{eff}} = \gamma_0 W$ over a cross-section πW^2 in a grain of size d. The macroscopic strain increment:

$$\delta\gamma_{0\text{macro}} = \frac{b_{\text{eff}} \cdot \pi W^2}{\pi W^2 \cdot d} = \frac{\gamma_0 W}{d}. \tag{72}$$

Step 2 (Eyring forward/backward rates).. Following Eyring [?], bias $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2$. Forward barrier $\Delta F_0 - \Delta w$; reverse $\Delta F_0 + \Delta w$:

$$\Gamma^{\pm} = \nu_0 \exp\left(-\frac{\Delta F_0 \mp \Delta w}{kT}\right). \tag{73}$$

Step 3 (factor out $e^{-\Delta F_0/kT}$).. Using $e^{a+b} = e^a \cdot e^b$:

$$\begin{aligned} e^{-(\Delta F_0 - \Delta w)/kT} &= e^{-\Delta F_0/kT} \cdot e^{+\Delta w/kT}, \\ e^{-(\Delta F_0 + \Delta w)/kT} &= e^{-\Delta F_0/kT} \cdot e^{-\Delta w/kT}. \end{aligned}$$

Step 4 ($e^x - e^{-x} = 2\sinh x$)..

$$\begin{aligned} \dot{N}_{\text{net}} &= \Gamma^+ - \Gamma^- \\ &= 2\nu_0 \sinh\left[\frac{(\tau-\tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT}. \end{aligned} \tag{74}$$

Step 5 (macroscopic rate).. Multiply by $\delta\gamma_{0\text{macro}} = \gamma_0 W/d$:

$$\boxed{\dot{\gamma_0} = \frac{2W\gamma_0\nu_0}{d} \sinh\left[\frac{(\tau-\tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT}.} \tag{75}$$

Identical to Padmanabhan et al. (1996, Eq. 8) [?].

7.3.1. Comparison: partition-function route

From $\ln \mathcal{Z} = N_{\text{sites}} \ln(1+e^{-h^{\text{eff}}/kT})$, differentiation with respect to $(\tau - \tau_0)\gamma_0 V_0/kT$ gives $N_{\text{sites}}\bar{n}$. In the dilute limit, this is a *forward-bias-only* expression — not equal to $2e^{-\Delta F_0/kT}\sinh(\Delta w/kT)$. The sinh rate equation is an Eyring *kinetic* result. Detailed balance: $\Gamma^+/\Gamma^- = e^{2\Delta w/kT}$. ✓

7.3.2. Connection to Padmanabhan et al. (1996)

With $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2$ and $(2W\gamma_0/d)\dot{w} = \dot{\gamma_0}$, Eq. (8) of [?] is recovered exactly. Including a lognormal disorder distribution $f(\Delta F_0)$:

$$\begin{aligned} \dot{\gamma_0} = \frac{2W\gamma_0\nu_0}{d} &\int_0^\infty f(\Delta F_0) \\ &\times \sinh\left[\frac{(\tau-\tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT}\,\mathrm{d}(\Delta F_0). \end{aligned} \tag{76}$$

Universal lognormal parameter $\bar{A} = 5.6021$ [?]; CV= 14.8%; range [0.28, 0.52] eV.

7.4. OSTZ interaction as the origin of Taylor hardening

When $z\bar{n}J_0$ is non-negligible, the effective threshold stress increases:

$$\tau_0^{\text{eff}} = \tau_0 + \frac{z\bar{n}J_0}{\gamma_0 V_0}. \tag{77}$$

With $\bar{n} = \pi W^2 \rho_{\text{disl}}$ and $J_0 \approx G\gamma_0^2 V_0^2/[4\pi(1-\nu)W^3]$, substituting $V_0 = (2\pi/3)W^3$:

$$\tau_0^{\text{eff}} - \tau_0 = \frac{z\gamma_0 G\pi W^2}{6(1-\nu)}\,\rho_{\text{disl}}. \tag{78}$$

OSET predicts hardening *linear* in ρ_{disl} (stage-I regime). The classical Taylor $\sqrt{\rho}$ law [?] requires the additional geometric forest-obstacle argument ($\tau_c = Gb/L$, $L \propto \rho^{-1/2}$) and is not derivable from the OSTZ Ising Hamiltonian alone.

8. Why OSET is More Fundamental

8.1. Logical priority theorem

Theorem. *OSET$\Rightarrow$dislocation theory, but not conversely.*

($\Rightarrow$): Sections ??–?? derive the Volterra field with quantised $b = N_c\gamma_0 W$.

($\not\Leftarrow$): Dislocation theory cannot describe (a) single OSTZ events; (b) OSTZ excitations in amorphous matter; (c) GBS without a DSC lattice; (d) the nanocrystal-to-glass transition. □

8.2. Ontological hierarchy

$$\text{OSTZ} \xrightarrow{N=1} \text{STZ/GBS} \xrightarrow{N=N_p} \text{partial disln} \xrightarrow{N=N_c} \text{full disln} \xrightarrow{N\gg N_c} \text{slip band}$$

Dislocations occupy one branch of the OSTZ tree. OSET is the root.

Table 5: Classical dislocation theory vs. OSET.

Property	Classical	OSET
Entity	Topological line	Eshelby spheroid
Lattice	Required	Not required
Materials	Crystals	All
Core stress	$\to\infty$	$\le G\gamma_0$
Core energy	Fitted αGb^2	Eq. (??)
Core width	Fitted	$\zeta = W$
Burgers b	Input	$N_c\gamma_0 W$
Thermal act.	Appended	Intrinsic
τ_P	W fitted	Eq. (??)
τ^*	Frenkel sep.	Eq. (??)
SFE	Fitted	Eq. (??)
GB plast.	DSC needed	Natural

9. OSET Across Material Classes

Perfect crystal ($\Delta F_0 \gg kT$). OSTZs are rare; act in clusters of N_c; recover Orowan equation.

Grain boundary / superplastic. OSTZs at GB free-volume sites; $N \ll N_c$; rate equation = Padmanabhan et al. [?] Eq. (39) directly.

Metallic glass (no N_c). $N = 1$ STZ picture [? ?]. At low T: localised deformation (shear bands = OSTZ percolation paths). At high T: homogeneous flow.

Nanocrystal near the glass transition ($d \to d_0$). As $d \to d_0 = 2\sqrt{6}W$, the threshold $\tau_0 \to 0$ and $N_c^{\text{eff}}(d) \to 1$: glass-like behaviour. OSET naturally spans the nanocrystal-to-glass transition.

10. Novel Predictions

P1: Core width $\zeta = W$ (testable by HAADF-STEM imaging of dislocation cores).

P2: Core energy without fitting. $E_{\text{core}}/\text{length} = \pi\beta_1 b\gamma_0 GW/6$; Cu: 0.124 eV/Å (DFT: 0.05–0.15 eV/Å).

P3: SFE scales as $\gamma_0^2 GW \approx \gamma_0^2 Gb$, explaining $\gamma_{\text{SF}} \propto Gb$ across FCC metals.

P4: Linear hardening $\tau_0^{\text{eff}} \propto \rho_{\text{disl}}$ (Eq. (??)) at low T; decreases with T through $\gamma_0(T)$ and $J_0 \propto G(T)$.

P5: Sub-dislocation displacement bursts $b_{\text{eff}} \approx 0.1b$, accessible with picometer-precision AFM-based nanoindentation.

11. Unified Rate Equation

Combining the OSTZ partition function (§??) with disorder-field theory (activation energy variance σ_F^2) and crystalline crossover Θ:

$$\dot{\gamma}_0 = \frac{2W\bar{\gamma}_0\nu_0}{d}\sinh\left[\frac{(\tau-\tau_0^*)\bar{\gamma}_0 V_0}{2kT}\right] \times \exp\left(-\frac{\Delta\bar{F}_0}{kT} + \frac{\sigma_F^2}{2(kT)^2}\right)\Theta(N_c, T, d), \quad (79)$$

where $\tau_0^* = \tau_0\sqrt{1+\rho_{G\gamma_0}\sigma_G\sigma_{\gamma 0}}$ and

$$\Theta = \begin{cases} 1 & \text{glass/GB regime} \\ N_c^{-1}\sum_{N=1}^{N_c} N\, e^{-E_c(N)/kT} & \text{crystal} \end{cases} \quad (80)$$

In the crystal limit $\Theta \to N_c$ (Orowan); in the glass/GB limit $\Theta \to 1$ (Padmanabhan et al. [?], Eq. (39)).

12. Validation

12.1. Validation against Padmanabhan et al. (1996)

T1 (Eshelby tensor): $\beta_1 = 0.446$, $\beta_2 = 0.889$. Exact.

T2 (Canonical): $\Delta F_0 = 0.405$ vs. 0.38 eV (6%).

T3 ($W = 2.5b$): $\delta V_{\text{free}} = \varepsilon_0 V_0 \approx 0.1\Omega$/GB atom; confirms Wolf MD [?].

T4 ($N_c = 4$): $b/(\gamma_0 W) = 0.3/(0.1\times 0.75) = 4.0$ exactly; confirmed by experiment in Zn–22Al [?].

T5 (Rate equation): Al-12Si at 773 K, $d = 10$ μm, $\tau - \tau_0 = 5$ MPa: $\Delta w = 2.2\times 10^{-22}$ J= 0.00138 eV; $\sinh(0.0207) = 0.0207$; $e^{-5.71} = 0.00330$; $\dot{\gamma}_0 = 0.16\ \text{s}^{-1}$ (superplastic window 10^{-3}–$10^{-1}\ \text{s}^{-1}$).

T6 (Threshold stress scaling $\tau_0 \propto L^{-1/2}$): Al-33Cu ratio predicted 2.0, measured 1.8–2.1.

T7 (Lognormal): $\bar{A} = 5.6021$, CV= 14.8%, range $[0.28, 0.52]$ eV.

Table 6: Predictions validated against Padmanabhan et al. (1996).

Claim	Pred.	Agr.
$\beta_1 = 0.446$	$1 - 2 \times 0.2772$	Exact
$\beta_2 = 0.889$	$(4/9)(4/3)/(2/3)$	Exact
$\gamma_0 = 0.10$	Wolf MD, bubble-raft	Exact
$\varepsilon_0 = 0.05$	$\delta V \approx 0.1\Omega$/atom	Exact
$N_c = 4$	4.0 derived	Exact (expt.)
$\Delta F_0 = 0.38$ eV	0.405 eV	6%
Rate eq. sinh	Eyring kinetics	Exact
$\tau_0 \propto L^{-1/2}$	ratio 2.0	$\leq 10\%$
$Q \sim$ 80–120 kJ/mol	90–120	$\leq 30\%$

12.2. Post-1996 literature

L1 Sripathi & Padmanabhan [?]: 33 systems, $R = 0.89$–1.00.

L2 β_1 convention reconciled: $0.446 \times 0.01 = 0.00446 \approx 1.779 \times 0.0025 = 0.00445$.

L3 $\gamma_0 = 0.10$: Wolf MD [?] (0.08–0.12), bubble-raft, Argon [?] (0.08–0.14), Falk–Langer [?] (0.10). Four independent methods.

L4 BMGs [?]: 8 systems, $\dot{\varepsilon}_{\text{pred}}/\dot{\varepsilon}_{\text{exp}} = 1.1$–2.5.

L5 Nano-Ni [?]: 8×10^{-4} vs. 5×10^{-4}–2×10^{-3} s^{-1} (factor 2).

L6 High-strain-rate superplasticity [?]: 8 alloy/composite systems; all within 2×.

L9 Peierls–Nabarro [?]: Fe exact; Cu within 2×.

L10 Lorentzian [?]: Al at $\rho = 10^{13}$ m^{-2}, $M = 0.16 < 0.5$ (Lorentzian regime).

12.3. Harisankar–Padmanabhan (2025): 41 systems

Table 7: OSET vs. Harisankar–Padmanabhan [?].

Quantity	OSET	Paper (mean)	Agr.
ε_0	0.05	0.049 ± 0.012	$< 2\%$
γ_0	0.10–0.12	0.085 ± 0.020	15%
$\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2$	0.0067	0.0057	15%
$Q/\Delta F_0$	1	≈ 4	see text
γ_B (J/m^2)	$E_0\theta_m$	0.30–1.70	4.5×

Key findings: (1) ε_0 predicted to $< 2\%$; (2) γ_0 mean 0.085 lies between the GB (0.10) and crystal (0.12) canonical values, consistent with temperature softening $\gamma_0(T) = 0.0827 + 1.342/T$; (3) $\Delta F_0 \approx Q/4$: the remaining three-quarters is non-elastic (GB migration, triple-junction accommodation, cooperative rearrangement of $N_a \approx 16$ boundary units); (4) γ_B to median 4.5× — a constant multiplicative offset showing no correlation with homologous temperature ($r = 0.00$), confirming correct Gb scaling.

13. Discussion

OSET establishes the ontological hierarchy: single OSTZ = elastic dipole = GBS/STZ event; N-chain = Peierls–Nabarro dislocation; $N = N_c$ = lattice dislocation.

Strengths.. FCC noble/transition metals (Cu, Ni, Au, Ag): SFE (5% for Cu), Peierls stress (within 2×), core energy — all without free parameters. GB/superplastic rate equation validated across 41 material systems.

Limitations.. (i) Al: anomalously high SFE requires $\gamma_0 \approx 0.30$ (NFE band structure). (ii) BCC metals: Peierls stress underpredicted 10–100× (BCC core has $W < b$; using $W = 0.2b$ recovers experiment). (iii) $Q/\Delta F_0 \approx 4$: explained by mesoscopic $N_a \approx 16$ ($Q = N_a\Delta F_0$) plus lognormal tail convolution; the elastic OSTZ barrier is 25% of the total.

14. Conclusions

1. OSTZ is lattice-free, non-singular, thermally activated. Eshelby parameters $\beta_1 = 0.446$, $\beta_2 = 0.889$ are derived rigorously from I_1, I_{13} (Sections ??–??).
2. Dislocations emerge as OSTZ collectives: dipole Eq. (??), Peierls–Nabarro profile Eq. (??), non-singular chain stress Eq. (??); Volterra singularity only as $W \to 0$, Eq. (??).
3. Five dislocation observables are derived without adjustable parameters: τ^* (??), τ_P (??), E_{core} (??), τ_{FR} (??), γ_{SF} (??).
4. The OSTZ Hamiltonian (??) yields the sinh flow law (??) and linear hardening (??).
5. Validation: ε_0 within 2%, γ_0 within 15%, γ_B to median 4.5×, across 41 superplastic systems.
6. OSET⇒dislocation theory (not conversely): OSET is the more fundamental description (Table ??).

CRediT

A. Linda: Methodology, Formal analysis, Writing. **K.A. Padmanabhan:** Conceptualization, Supervision, Review.

Competing interests

None.

Data availability

Jupyter notebook with all derivations and numerical verifications accompanies this article.

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